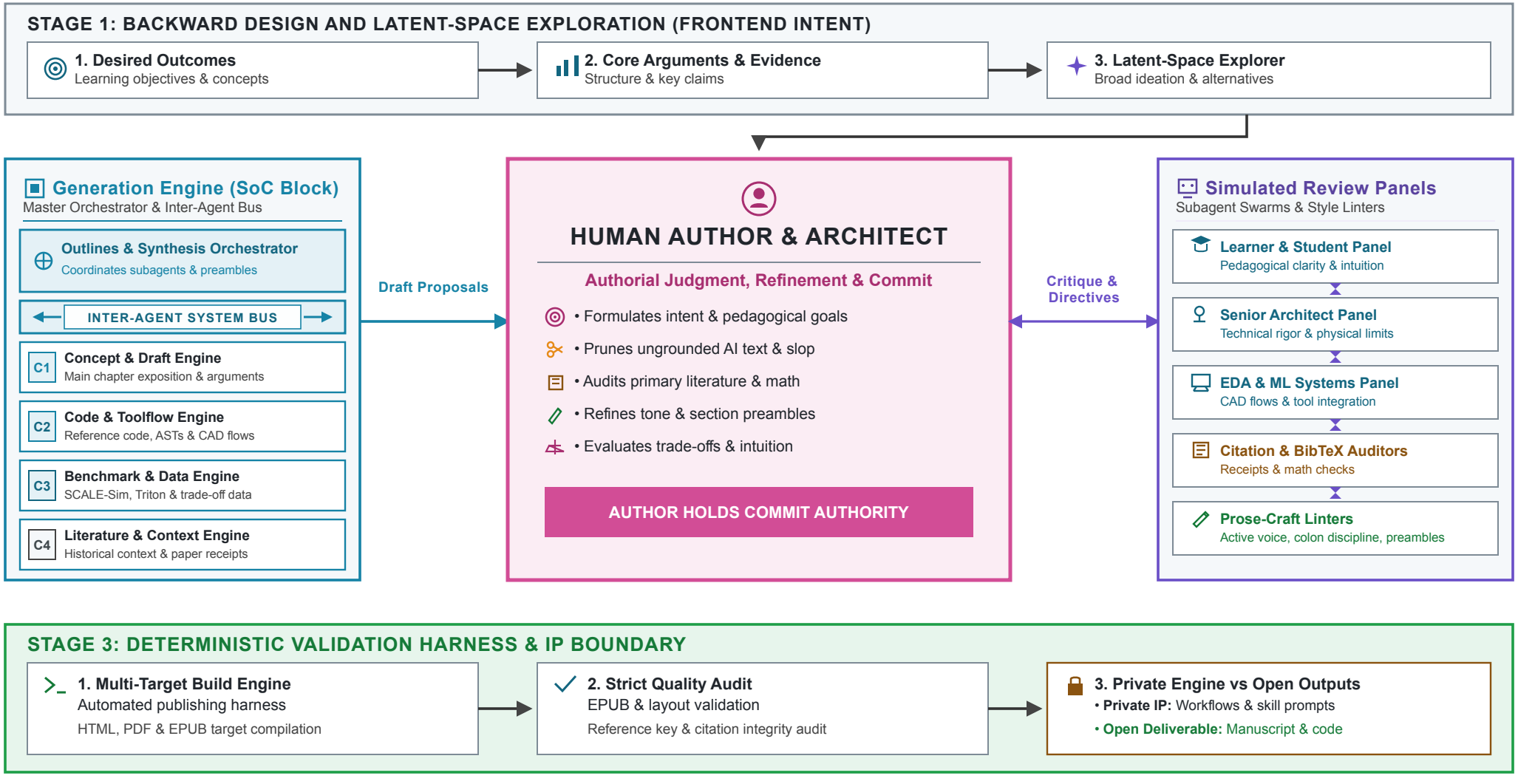

STAGE 1: BACKWARD DESIGN AND LATENT-SPACE EXPLORATION (FRONTEND INTENT)
1. Desired Outcomes
Learning objectives & concepts
2. Core Arguments & Evidence
Structure & key claims
3. Latent-Space Explorer
Broad ideation & alternatives
Generation Engine (SoC Block)
Master Orchestrator & Inter-Agent Bus
Outlines & Synthesis Orchestrator
Coordinates subagents & preambles
INTER-AGENT SYSTEM BUS
C1
Concept & Draft Engine
Main chapter exposition & arguments
C2
Code & Toolflow Engine
Reference code, ASTs & CAD flows
C3
Benchmark & Data Engine
SCALE-Sim, Triton & trade-off data
C4
Literature & Context Engine
Historical context & paper receipts
Draft Proposals
HUMAN AUTHOR & ARCHITECT
Authorial Judgment, Refinement & Commit
• Formulates intent & pedagogical goals
• Prunes ungrounded AI text & slop
• Audits primary literature & math
• Refines tone & section preambles
• Evaluates trade-offs & intuition
AUTHOR HOLDS COMMIT AUTHORITY
Critique & Directives
Simulated Review Panels
Subagent Swarms & Style Linters
Learner & Student Panel
Pedagogical clarity & intuition
Senior Architect Panel
Technical rigor & physical limits
EDA & ML Systems Panel
CAD flows & tool integration
Citation & BibTeX Auditors
Receipts & math checks
Prose-Craft Linters
Active voice, colon discipline, preambles
STAGE 3: DETERMINISTIC VALIDATION HARNESS & IP BOUNDARY
1. Multi-Target Build Engine
Automated publishing harness
HTML, PDF & EPUB target compilation
2. Strict Quality Audit
EPUB & layout validation
Reference key & citation integrity audit
3. Private Engine vs Open Outputs
• Private IP: Workflows & skill prompts
• Open Deliverable: Manuscript & code